

SUMMARY

Backend engineer with 7+ years building distributed systems in Java 17-21/Spring Boot and Python/FastAPI. Sole backend developer at Smart Microwave Sensors in a full-stack role, and the team's final reviewer for all backend and frontend code and design decisions. Delivered 20+ production microservices at Walmart, built event-driven pipelines with Kafka, ran services on Kubernetes, and owned workflow automation with Camunda 7. Built a multi-repo AI developer agent using LangGraph, FastMCP, and Qdrant-backed RAG. Strong on system design, security (JWT/OAuth2), observability, and CI/CD. Looking for senior backend or AI-backend roles. Open to relocation across EU, UK, Australia/NZ, and UAE.

TECHNICAL SKILLS

Languages	Java 17-21 • Python 3.x • JavaScript (ES6+) • TypeScript • Shell Scripting
Backend	Spring Boot • FastAPI • Hibernate • REST APIs • GraphQL • Microservices
AI & LLM	LLM Integration • RAG • LangChain • LangGraph • FastMCP (MCP) • pgvector • Qdrant • Tree-sitter (AST) • Prompt Engineering • AI Agents • LangSmith • Groq • OpenAI Embeddings
Cloud & DevOps	GCP (BigQuery, GCS, Pub/Sub) • Kubernetes (Helm) • Docker • Docker Compose • GitHub Actions • GitLab CI • Maven • Gradle • Linux/Ubuntu • Nginx • Railway • Vercel
Messaging	Apache Kafka • MQTT • Event-Driven Architecture • Async Processing • Celery • Redis
Security	JWT • OAuth2 • Sidecar-based Service Auth • Spring Security
Workflow/BPM	Camunda 7 (Certified) • BPMN 2.0 (Certified) • Process Automation
Databases	PostgreSQL • pgvector • MongoDB • Redis • BigQuery • SQL Server
Observability	Grafana • Prometheus • LangSmith • Distributed Tracing • SLO/SLA • Incident Response
Practices	System Design • DDD • Agile/Scrum • TDD • Mentoring • Technical Leadership • Code Review • On-Call • OpenAPI/Swagger • Postman • Mockoon

WORK EXPERIENCE

Senior Full Stack Engineer (Backend-Focused) — Smart Microwave Sensors GmbH • Braunschweig, Germany

2025 – Present

Apr

- Sole backend developer on the team in a full-stack role; own backend architecture and API design, and handle major frontend development alongside the frontend engineers.
- Own PR review and final architecture/API design decisions for the backend; onboarded and mentored a new hire on the backend stack and workflows.
- Final reviewer for all frontend code and design changes, including sign-offs requested by the team lead; mentored two frontend engineers on code quality and design.
- Own API contract design; gather requirements from downstream teams in daily syncs and translate them into REST API contracts before implementation.
- Coordinate with the embedded Linux team on Yocto-level changes for solutions that can't run directly on-device due to security or platform constraints.
- Built a browser-based MQTT traffic analysis tool (Python, React) extending MQTT Explorer with configurable-frequency message polling for real-time traffic analysis.
- Built the FastAPI backend from scratch: MQTT ingestion from 100+ road intersections, REST APIs to a React dashboard, sub-200ms latency via MQTT broker tuning.
- Owned the graph component service end to end: requirements gathering, POC, and the Recharts dashboard showing live per-intersection traffic data.
- Built the D3.js/SVG map layer on Google Maps and Bing Maps for real-time sensor positions, lane markings, and detection zones.
- Re-architected Mockoon so only explicitly listed APIs return mocked data (default behavior wasn't compatible with the existing frontend architecture), cutting frontend blocking time by roughly 2 sprints per feature cycle.
- Follow TDD in the FastAPI backend, writing unit and integration tests with pytest; regression tests run as part of the CI pipeline before every deploy.
- Containerized services with Docker; set up a CI pipeline for automated tests and deployment on every merge to main.

Software Engineer 3 – Backend Platform — Walmart Global Tech • Bengaluru, India

Jan 2022 – Mar 2025

- Owned 20+ Java 17/Spring Boot microservices across supply chain, pricing, and promotions, serving millions of daily requests on Walmart.com.
- Queried pricing data from the pricing team's GraphQL API as part of microservice integrations.
- Built Wrangler, a Java/Camunda 7 wrapper adopted by 10+ teams; reduced workflow integration time from 2-3 weeks to a few days.
- Built Execution Services, a Java PaaS between Camunda and external task processors, adopted by 10+ downstream teams for polling, execution, callbacks, and retry logic.
- Replaced synchronous HTTP calls across 15-20+ service endpoints with Kafka-based async messaging, removing the main throughput bottleneck.

- Implemented JWT/OAuth2-based authentication and sidecar containers for upstream/downstream service-to-service auth, following established platform security patterns.
- Set up GitHub Actions across 20+ services for automated tests, quality gates, and blue-green deploys; built Grafana/Prometheus dashboards that cut MTTD on P1 incidents by around 40%.
- Practiced TDD across Java/Spring Boot microservices, writing unit and integration tests with JUnit; regression suites ran as GitHub Actions quality gates before every deploy.
- Deployed and operated services on Kubernetes: wrote Helm charts, scaled pods up and down based on load, and debugged production issues like OOM kills and pod restarts via logs.
- Built Grafana dashboards on Kubernetes/Prometheus infra metrics tracking error counts and response counts across services.
- Authored API/backend and Grafana dashboard documentation (Prometheus and SQL query data) on Confluence, used as reference material by other engineering teams.
- Built Cost Driver, a hackathon project forecasting grocery item prices from ingredient costs to maximize profit margins; presented at Walmart's hackathon showcase.
- Wrote BigQuery queries to pull retail forecast data from the data science team's datasets and fed workflow-generated data back for their forecasting models.
- Used GCP Cloud Storage for storing and retrieving workflow-related artifacts in the Camunda-based services.
- 3 years on the senior on-call rotation; resolved or led resolution on dozens of P1 incidents and authored the domain runbooks. Multiple Impact Driver and Firefighter awards for P1 resolution speed.

Teaching Assistant – Algorithms (Part-Time) — **Coding Ninjas** • Remote Aug 2021 – Dec 2021

- Taught DSA in Python to 100+ students and working professionals. Received a 4.9/5 rating.

Software Engineering Specialist – Backend — **Dassault Systèmes** • Pune, India Jan 2020 – Jan 2022

- Owned the CRS Framework REST APIs, used in production by Ericsson, Melexis, and Framatome.
- Developed TRM, replacing IBM DOORS for requirements traceability across PLM lifecycle data; set up the team's first GitLab CI pipelines.

Associate IT Developer — **India Medtronic Pvt. Ltd.** • Bengaluru, India Feb 2019 – Jan 2020

- Built NCG, a Python/Flask tool auto-generating Cisco Switch 3850-48 configs from templates, replacing a manual process across Medtronic's global network infrastructure.

KEY PROJECTS

Wrangler + Execution Services | Walmart Global Tech

Wrangler: Java/Spring Boot wrapper over Camunda 7, adopted by 10+ teams; cut integration time from weeks to days. Execution Services: Java PaaS connecting Camunda to external task processors with automated polling, execution, and failure recovery at retail scale.

Stack: Java 17 • Spring Boot • Camunda 7 • PostgreSQL • Kafka • GitHub Actions

COMHUB – Real-Time Traffic Monitoring System | Smart Microwave Sensors

Traffic monitoring system built from a blank repo to production, solo: FastAPI + MQTT backend, React + Recharts + D3.js frontend, sub-200ms latency from 100+ sensors.

Stack: FastAPI • Python • MQTT • PostgreSQL • Docker • GitHub Actions

Multi-Repo AutoPR & Anti-Drift Agent | AI Developer Co-Pilot, Smart Microwave Sensors

- Architected an Autonomous Multi-Repo Agent: Engineered an agentic workflow using LangGraph state machines and FastMCP micro-servers to execute end-to-end feature development across microservices.
- Engineered Dual-Tier Codebase RAG: Combined Tree-sitter AST parsing and Qdrant to isolate active domain code from foreign API contracts (openapi.json, Protobufs), cutting prompt token usage by 95%.
- Enforced Pre-Push Pipeline Validation: Integrated a local Docker runner executing linters, type checks, and integration tests with an automated self-healing loop prior to GitHub PR creation.
- Built Zero-Token Anti-Drift Daemon: Developed a daily background service leveraging deterministic schema diffing (git diff, openapi-diff) to detect cross-service API drift and generate automated draft PRs.

Stack: Python • LangGraph • FastMCP (MCP) • Qdrant • Tree-sitter • Docker • Jira API • GitHub API

EDUCATION

B.Tech – Computer Science & Engineering — **Harcourt Butler Technical University** • Kanpur, India 2013 – 2017

CERTIFICATIONS & RECOGNITION

- Camunda 7 Java Developer Certified • BPMN 2.0 Certified • Google Cloud Digital Leader
- 15 Walmart badges • Multiple Impact Driver & Firefighter awards for P1 resolution
- Certificate of Excellence – DSA in Python, Coding Ninjas (Top 10%)

LANGUAGES

English (Fluent) • Hindi (Native) • German (Learning – A1)